

Course Name and Code	<i>Data Security - ITI410</i>
Number	<i>1</i>
Title	<i>OpenSSL</i>
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Revision by	
Classes	<i>C14 and C15</i>
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Resources	<i>Cryptography Techniques</i>

OBJECTIVES

The assignment aims at enforcing the cryptography techniques. It helps the student in acquiring the necessary knowledge in understanding cryptography applications and in securing data using encryption techniques.

Objectives can be summarized as:

- Understanding cryptography applications
- Encrypting and decrypting using DES and RSA
- Hashing using SHA1

The assignment aims at making the student familiar with certain basic and advanced security techniques; essentially asymmetric cryptography and certificates. You will be able to employ these services in a scenario of secure message exchange.

This assignment is based on the OpenSSL toolkit. OpenSSL is a widely-used open-source, cryptography API.

ASSIGNMENT DETAILS

1. Purpose

The purpose is to master OpenSSL, an open source toolkit providing general purpose cryptography features. It aims at mastering the following issues:

1. Creating Private/Public keys
2. Encrypting/Decrypting using symmetric cryptography algorithms.
3. Encrypting/Decrypting using asymmetric cryptography algorithms.
4. Hashing content.
5. Signing content.

To achieve these goals, the assignment asks the student to build a Certification Authority (CA). The CA responsibilities are:

- Creating its own certificate
- Creating certificates for customers

The customers can exchange encrypted and signed data using their certificates.

2. Functionalities

2-1. Basic Symmetric and Asymmetric Practices

You will practice on using OpenSSL to handle some basic symmetric and asymmetric functions.

1. Create a text file called "message.txt" containing the full name, ID and class of each one of the group members.
2. Write a command line in a batch file called "Gen_DES3_Key.bat" to generate a DES3 key coded in base64. Run the command and save the key in a file called "DES3_key.txt".
3. Write a command line in a batch file called "Encrypt_DES3.bat" to encrypt the file "message.txt" using the DES3 algorithm and the key stored in "DES3_key.txt". Run the command and save the result in a file called "Encrypted_DES3.txt".
4. Write a command line in a batch file called "Decrypt_DES3.bat" to decrypt the file "Encrypted_DES3.txt" using the DES3 algorithm and the key stored in "DES3_key.txt". Run the command and save the result in a file called "Decrypted_DES3.txt".
5. Write a command line in a batch file called "Generate_RSAKeys.bat" to generate RSA private/public keys of 2048-bits length. Run the command and save the RSA keys in a file called "RSAKeys.pem".
6. Write a command line in a batch file called "Extract_RSAPublic.bat" to extract the RSA public key from the RSA keys file "RSAKeys.pem". Run the command and save the public key in a file called "RSA_PublicKey.pem".
7. Write a command line in a batch file called "Encrypt_RSA_Public.bat" to encrypt the file "DES3_key.txt" using the RSA public key. Run the command and save the result in a file called "Encrypted_RSA_Public.txt".
8. Write a command line in a batch file called "Decrypt_RSA_Private.bat" to decrypt the file "Encrypted_RSA_Public.txt" using the RSA private key. Run the command and save the result in a file called "Decrypted_RSA_Private.txt".

2-2. Certification

You will create a Certification Authority (CA) and two MIME clients. You need first to create a subfolder called "newcerts", a text file called "serial.txt" containing the text "01" and an empty text file called "index.txt".

9. Write a command line in a batch file called "Generate_CAKeys.bat" to generate the public/private key pair of the email Certification Authority (MIME CA) in PEM format. Run the command and save the result in a file called "ca.key".
10. Write a command line in a batch file called "Create_CACert.bat" to create the auto-signed certificate of the CA in PEM format using the configuration file "openssl.txt" provided with the assignment. Run the command and save the result in a file called "ca.pem".
11. Write a command line in a batch file called "Visualize_CACert.bat" to create a readable version out of "ca.pem" to visualise the content of the auto-

- signed certificate of the CA. Run the command and save the result in a file called "ca_cert.txt".
12. Write the needed command lines in a batch file called "Generate_ClientKeys.bat" to generate the public/private key pairs for two email clients (MIME clients) "C1" and "C2" in PEM format. Run the commands and save the results in two files called "C1.key" and "C2.key" respectively.
 13. Write the needed command lines in a batch file called "Generate_ClientRequests.bat" to generate a certificate request for each client using the configuration file "openssl.txt" provided with the assignment. Run the commands and save the results in two files called "C1.crs" and "C2.crs" respectively.
 14. The certificate requests should be sent to the CA to be signed, which should generate the client certificates. Taking the role of the CA, write the needed command lines in a batch file called "Sign_ClientRequests.bat" to sign the requests "C1.crs" and "C2.crs" of the two MIME clients, in order to generate their respective identity certificates. Run the commands and save the results in two files called "Signed_C1_CRS.ca" and "Signed_C2_CRS.ca" respectively.

2-3. Sending Secure Messages

Client "C1" will send a message to client "C2" securely. Client "C1" will encrypt and sign his/her message and send it to client "C2".

15. Choose a password of 12 characters at least and store in a file called "pwd.txt".
16. Write a command line in a batch file called "Encrypt_AES.bat" to encrypt the message "message.txt" using the password stored in "pwd.txt" and the AES algorithm. Run the commands and save the results in a file called "message.enc".
17. Write a command line in a batch file called "Encrypt_pwd.bat" to encrypt the password stored in "pwd.txt" using the public key of the client "C2" (his/her certificate). Use the command "smime" for this purpose. Run the command and save the result in a file called "pwd.enc".
18. Write a command line in a batch file called "Generate_HashC1.bat" to generate a hash value out of the message "message.txt" using the MD5 algorithm. Run the command and save the result in a file called "message_hash.txt".
19. Write a command line in a batch file called "Sign_Message.bat" to sign the hash value stored in "message_hash.txt" using your private key. Use the command "smime" for this purpose. Run the commands and save the results in a file called "message.sig".

2-4. Receiving Secure Messages

Client "C2" will receive the encrypted and signed message sent by client "C1". It will receive the encrypted message text "message.enc", the encrypted password "pwd.enc" and the message signature "message.sig". Client "C2" will decrypt and verify the received message.

20. Write a command line in a batch file called "Decrypt_Pwd.bat" to decrypt the encrypted password "pwd.enc" using the private key of client "C2". Use the command "smime" for this purpose. Run the command and save the result in a file called "pwd_dec.txt".
21. Write a command line in a batch file called "Decrypt_AES.bat" to decrypt the encrypted message "message.enc" using the password stored in the file "pwd_dec.txt" and the AES algorithm. Run the commands and save the result in a file called "message_dec.txt".
22. Write a command line in a batch file called "Generate_HashC2.bat" to generate a hash value out of the message "message_dec.txt" using the MD5 algorithm. Run the commands and save the result in a file called "message_dec_hash.txt".
23. Write a command line in a batch file called "Verify_Message.bat" to verify the signature stored in the file "message.sig" using the public key of the client "C1" (his/her certificate). Use the command "smime" for this purpose. Run the commands and save the result in a file called "message_verify_hash.txt".
24. Explain in a Word document called "Mail_Security.doc" the security concepts (Confidentiality, Integrity and Availability) that this secure mail exchange model achieves. Also identify if any vulnerabilities could exist.

DELIVERABLES

The student should deliver a one-folder package containing – in addition to the subfolder "newcerts", the text file "serial.txt", the text file "index.txt" and the configuration file "openssl.txt" – the following files:

1. message.txt
2. Gen_DES3_Key.bat
3. DES3_key.txt
4. Encrypt_DES3.bat
5. Encrypted_DES3.txt
6. Decrypt_DES3.bat
7. Decrypted_DES3.txt
8. Generate_RSAKeys.bat
9. RSAKeys.pem
10. Extract_RSAPublic.bat
11. RSA_PublicKey.pem
12. Encrypt_RSA_Public.bat
13. Encrypted_RSA_Public.txt
14. Decrypt_RSA_Private.bat
15. Decrypted_RSA_Private.txt
16. Generate_CAKeys.bat
17. ca.key
18. Create_CACert.bat
19. ca.pem
20. Visualize_CACert.bat
21. ca_cert.txt
22. Generate_ClientKeys.bat
23. C1.key
24. C2.key
25. Generate_ClientRequests.bat
26. C1.crs
27. C2.crs

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28.Sign_ClientRequests.bat
29.Signed_C1_CRS.ca
30.Signed_C2_CRS.ca
31.pwd.txt
32.Encrypt_AES.bat
33.    message.enc
34.Encrypt_pwd.bat
35.    pwd.enc
36.Generate_HashC1.bat
37.    message_hash.txt
38.Sign_Message.bat
39.    message.sig
40.Decrypt_Pwd.bat
41.    pwd_dec.txt
42.Decrypt_AES.bat
43.    message_dec.txt
44.Generate_HashC2.bat
45.    message_dec_hash.txt
46.Verify_Message.bat
47.    message_verify_hash.txt
48.    Mail_Security.doc
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ASSIGNMENT DELIVERY

1. Students can work in groups of four at maximum. Student names in each group should be communicated to the tutor prior to assignment delivery.

2. A ZIP package should be delivered. The file will be named as follows:
Assignment_ITI410_F11_<Student1ID>_<Student2ID>.zip

For example:

Assignment_ITI410_F11_mahmoud_13452_jameel_16272.zip
Assignment_ITI410_F11_mouna_17499.zip

3. The Assignment solution should be posted to the EXAMS system. The assignment question title on EXAMS is "BIT_ITI410_C11C12C13_F11_Assignment"
Assignment delivery through Email is NOT accepted.

ASSIGNMENT SOURCE AND RESOURCES

1. The assignment source will be sent to students by email and it can be downloaded from the SVU website.

2. The OpenSSL package to be used is "Win32OpenSSL-1_0_0e.exe" and it can be downloaded from:

http://www.slproweb.com/download/Win32OpenSSL-1_0_0e.exe

3. To read about OpenSSL go to the official website <http://www.openssl.org/>